

Observe that if we force $\tau = \gamma = 1$ and $\omega = 0$, then we recover the corresponding formulae given in Proposition 2. On the other hand, taking $\Sigma_1 = \Sigma_2 = \Sigma$ gives Proposition 3.

Proof. WLOG, we only consider the cases where $j = 1$.

Computing $r_1^{(1)}$. There is a 11×11 minimal linear pencil Q such that $ASRS^\top M_1 = Q^{-1}[1, 10]$ (zero-based indexing). We deduce that

$$r_1^{(1)} := \mathbb{E} \operatorname{tr} ASRS^\top M_1 \simeq d \cdot G_{1,10}, \quad (127)$$

where $G := (\operatorname{id} \otimes \mathbb{E} \bar{\operatorname{tr}})Q^{-1} \in \mathbb{C}^{11 \times 11}$. Moreover, $G_{1,10}$ is given by the following fixed-point equations

$$G_{1,10} = p_1 \gamma G_{2,2} G_{5,5} \bar{\operatorname{tr}} A \Sigma_1 K^{-1}, \quad (128)$$

$$\text{with } K := \gamma G_{2,2} L + \lambda I_d, \quad L := p_1 G_{5,5} \Sigma_1 + p_2 G_{8,8} \Sigma_2, \quad (129)$$

$$G_{5,5} = \frac{1}{1 + \phi \gamma G_{2,2} \bar{\operatorname{tr}} \Sigma_1 K^{-1}} = \frac{1}{1 + \psi G_{2,2} \bar{\operatorname{tr}} \Sigma_1 K^{-1}}, \quad (130)$$

$$G_{8,8} = \frac{1}{1 + \phi \gamma G_{2,2} \bar{\operatorname{tr}} \Sigma_2 K^{-1}} = \frac{1}{1 + \psi G_{2,2} \bar{\operatorname{tr}} \Sigma_2 K^{-1}}, \quad (131)$$

$$G_{2,2} = \frac{1}{1 + \bar{\operatorname{tr}} L K^{-1}}, \quad (132)$$

Then, one deduces that

$$\operatorname{tr} ASRS^\top M_1 \simeq d \cdot G_{1,10} = p_1 e_1 \tau \gamma \operatorname{tr} A \Sigma_1 K^{-1}. \quad (133)$$

Computing $r_1^{(4)}$. Here, the pencil Q is 20×20 and $AM_1 SRS^\top SRS^\top = -Q^{-1}[1, 13]/\lambda$. We deduce that

$$r_1^{(4)} := \mathbb{E} \operatorname{tr} AM_1 SRS^\top B SRS^\top \simeq -d \cdot G_{1,13}/\lambda, \quad (134)$$

where $G := (\operatorname{id} \otimes \mathbb{E} \bar{\operatorname{tr}})Q^{-1} \in \mathbb{C}^{20 \times 20}$. Moreover, $G_{1,13}$ is given by the following fixed-point equations

$$-G_{1,13} = p_1 \gamma \bar{\operatorname{tr}} A \Sigma_1 T K^{-2}, \quad \text{where} \quad (135)$$

$$T := \lambda(\tau^2 e_1 B + (e_1 G_{6,12} + \tau G_{3,15}) I_d) + p_2 \gamma \tau^2 (e_2 G_{3,15} - e_1 G_{9,18}) \Sigma_2, \quad (136)$$

$$G_{12,12} = G_{6,6} = \tau, \quad (137)$$

$$G_{3,15} = \frac{\phi e_1^2 G_{4,14}}{\lambda}, \quad (138)$$

$$G_{4,14} = -\lambda \gamma \bar{\operatorname{tr}} \Sigma_1 (\gamma \tau^2 (p_1 G_{3,15} \Sigma_1 + p_2 G_{9,18} \Sigma_2) - \lambda(\gamma \tau^2 B + G_{6,12} I_d)) K^{-2}, \quad (139)$$

$$G_{9,18} = \frac{\phi e_2^2 G_{10,17}}{\lambda}, \quad (140)$$

$$G_{10,17} = -\lambda \gamma \bar{\operatorname{tr}} \Sigma_2 (\gamma \tau^2 (p_1 G_{3,15} \Sigma_1 + p_2 G_{9,18} \Sigma_2) - \lambda(\gamma \tau^2 B + G_{6,12} I_d)) K^{-2}, \quad (141)$$

$$G_{6,12} = -\tau^2 G_{7,11}, \quad (142)$$

$$G_{7,11} = -\bar{\operatorname{tr}} (\gamma K_0^2 + \lambda(\lambda B - p_1 G_{3,15} \Sigma_1 - p_2 G_{9,18} \Sigma_2)) K^{-2}, \quad (143)$$

We deduce that $G_{3,15} = -v_1$, $G_{9,18} = -v_2$, and $G_{6,12} = \omega$, where $v_1, v_2, \omega \geq 0$ solve the following fixed-point equations

$$\begin{aligned} v_1 &= \phi \gamma e_1^2 \bar{\operatorname{tr}} \Sigma_1 (\gamma \tau^2 (p_1 v_1 \Sigma_1 + p_2 v_2 \Sigma_2) + \lambda(\gamma \tau^2 B + \omega I_d)) K^{-2} \\ &= \psi e_1^2 \bar{\operatorname{tr}} \Sigma_1 (\gamma \tau^2 L + \lambda \omega I_d) K^{-2}, \\ v_2 &= \phi \gamma e_2^2 \bar{\operatorname{tr}} \Sigma_2 (\gamma \tau^2 (p_1 v_1 \Sigma_1 + p_2 v_2 \Sigma_2) + \lambda(\gamma \tau^2 B + \omega I_d)) K^{-2} \\ &= \psi e_2^2 \bar{\operatorname{tr}} \Sigma_2 (\gamma \tau^2 L + \lambda \omega I_d) K^{-2}, \\ \omega &= \tau^2 \bar{\operatorname{tr}} (\gamma K_0^2 + \lambda(\lambda B + p_1 v_1 \Sigma_1 + p_2 v_2 \Sigma_2)) K^{-2} = \tau^2 \bar{\operatorname{tr}} (\gamma K_0^2 + \lambda L) K^{-2}, \end{aligned}$$

with $L := p_1 v_1 \Sigma_1 + p_1 v_2 \Sigma_2 + \lambda B$. Putting everything together then gives

$$r_j^{(4)} \simeq -\frac{d \cdot G_{1,13}}{\lambda} = p_1 \gamma \operatorname{tr} A \Sigma_1 \tilde{T} K^{-2}, \text{ where}$$

$$\tilde{T} := T/\lambda = \tau^2 e_1 B + (e_1 \omega - \tau v_1) I_d + p_2 \gamma \tau^2 (e_1 u_2 - e_2 u_1) \Sigma_2 =: D_1.$$

Computing $r_1^{(3)}$. The matrix of interest $AM_1 SRS^\top BSRS^\top M_1$ admits a minimal linear pencil Q of size 21×21 , such that the formal equals to $Q^{-1}[1, 20]$. It follows that

$$r_1^{(3)} := \mathbb{E} \operatorname{tr} AM_1 SRS^\top BSRS^\top M_1 \simeq d \cdot G_{1,20}, \quad (144)$$

where $G := (\operatorname{id} \otimes \mathbb{E} \bar{\operatorname{tr}}) Q^{-1} \in \mathbb{C}^{21 \times 21}$. The fixed-point equations defining $G_{1,20}$ are

$$G_{1,20} = p_1 \bar{\operatorname{tr}} A \Sigma_1 (T/\lambda) K^{-2}, \text{ where}$$

$$T := p_1 G_{3,3}^2 \gamma (\gamma \tau^2 (\lambda B - p_2 G_{9,18} \Sigma_2) + \lambda G_{6,12} I_d) \Sigma_1 - G_{3,15} (\gamma \tau p_2 G_{9,9} \Sigma_2 + \lambda I_d)^2,$$

$$G_{3,3} = e_1,$$

$$G_{9,9} = e_2,$$

$$G_{6,12} = \omega,$$

$$G_{3,15} = -v_1,$$

$$G_{9,18} = -v_2.$$

Putting things together gives

$$r_1^{(3)} \simeq d \cdot G_{1,20} = \operatorname{tr} A \Sigma_1 \tilde{T} K^{-2},$$

where $\tilde{T} := T/\lambda = \gamma p_1 e_1^2 (\gamma \tau^2 (B + p_2 u_2 \Sigma_2) + \omega I_d) \Sigma_1 + u_1 (\gamma \tau p_2 e_2 \Sigma_2 + \lambda I_d)^2 =: C_1$, which completes the proof. $\square$

F.2 Proof of Theorem 2

This follows directly from Proposition 3 and the computations in Section C.2.

F.3 Recovering Theorem 1 from Theorem 2

Indeed, we have

$$\omega' \rightarrow 0, \quad \theta \rightarrow \kappa, \quad u \rightarrow \frac{\phi I_{2,2}(\kappa)}{1 - \phi I_{2,2}(\kappa)} = \frac{\operatorname{df}_2(\kappa)/n}{1 - \operatorname{df}_2(\kappa)/n},$$

for any regularization strength $\lambda > 0$, where κ is as defined in equation (7). Refer to Lemma 1. Plugging these limits into the formulae provided in Theorem 2 then recovers Theorem 1.

G Phase-Diagram for Random Projections Model

G.1 The General Regularized Case

Lemma 1. *The scalars u and ω' which appear in Theorem 2, and described in Definition 2, solve the following pair of linear equations*

$$u = \phi I_{2,2}(\theta)(1 + u) + \phi I_{1,2}(\theta) \omega',$$

$$\gamma \omega' = I_{2,2}(\theta) \omega' + \theta^2 I_{1,2}(\theta)(1 + u). \quad (145)$$

Furthermore, the solutions can be explicitly represented as

$$u = \frac{\phi z}{\gamma - \phi z - I_{2,2}(\theta)}, \quad \omega' = \frac{\theta^2 I_{2,2}(\theta)}{\gamma - \phi z - I_{2,2}(\theta)}, \quad (146)$$

where $z = I_{2,2}(\theta)(\gamma - I_{2,2}(\theta)) + \theta^2 I_{1,2}(\theta)^2$.

In particular, in the limit $\gamma \rightarrow \infty$, it holds that

$$\theta \simeq \kappa, \quad \omega' \rightarrow 0, \quad u \simeq \frac{\phi I_{2,2}(\kappa)}{1 - \phi I_{2,2}(\kappa)} \simeq \frac{\operatorname{df}_2(\kappa)/n}{1 - \operatorname{df}_2(\kappa)/n}, \quad (147)$$

where $\kappa > 0$ is as defined in (7).

Proof. The equations defining these are

$$u = \psi e^2 \bar{\text{tr}} \Sigma (\gamma \tau^2 L' + \omega I_d) K^{-2}, \quad (148)$$

$$\omega = \tau^2 \bar{\text{tr}} (\gamma \omega K_0^2 + \lambda^2 L') K^{-2}, \quad (149)$$

where $K_0 = e\Sigma$, $K = \gamma\tau K_0 + \lambda I_d$, and $L' := u\Sigma + B$. Further, since $B = \Sigma$, we have $L' = (1 + u)\Sigma$. Now, we can rewrite the previous equations like so

$$u = \psi e^2 \bar{\text{tr}} \Sigma (\gamma \tau^2 (1 + u)\Sigma + \omega I_d) K^{-2} = \phi \gamma^2 \tau^2 e^2 (1 + u) \bar{\text{tr}} \Sigma^2 K^{-2} + \phi \gamma e^2 \omega \bar{\text{tr}} \Sigma K^{-2},$$

$$\omega = \tau^2 \bar{\text{tr}} (\gamma \omega e^2 \Sigma^2 + \lambda^2 (1 + u)\Sigma) K^{-2} = \gamma \tau^2 e^2 \omega \bar{\text{tr}} \Sigma^2 K^{-2} + \lambda^2 \tau^2 (1 + u) \bar{\text{tr}} \Sigma K^{-2}.$$

This can be equivalently written as

$$u = \phi(1 + u) \gamma^2 \tau^2 e^2 \bar{\text{tr}} \Sigma^2 K^{-2} + \phi \omega' \gamma^2 \tau^2 e^2 \bar{\text{tr}} \Sigma K^{-2}, \quad (150)$$

$$\gamma \omega' = \omega' \gamma^2 \tau^2 e^2 \bar{\text{tr}} \Sigma^2 K^{-2} + (1 + u) \lambda^2 \bar{\text{tr}} \Sigma K^{-2}. \quad (151)$$

Now, observe that

$$\tau^2 e^2 \bar{\text{tr}} \Sigma^2 K^{-2} = \bar{\text{tr}} \Sigma^2 (\Sigma + \theta I_d)^{-2} / \gamma^2 = I_{2,2}(\theta) / \gamma^2, \quad (152)$$

$$\tau^2 e^2 \bar{\text{tr}} \Sigma K^{-2} = \bar{\text{tr}} \Sigma (\Sigma + \theta I_d)^{-2} / \gamma^2 = I_{1,2}(\theta) / \gamma^2, \quad (153)$$

$$\lambda^2 \bar{\text{tr}} \Sigma K^{-2} = \theta^2 \bar{\text{tr}} \Sigma (\Sigma + \theta I_d)^{-2} = \theta^2 I_{1,2}(\theta), \quad (154)$$

$$e^2 \bar{\text{tr}} \Sigma K^{-2} = \bar{\text{tr}} \Sigma (\Sigma + \theta I_d)^{-2} / (\gamma \tau)^2 = I_{1,2}(\theta) / (\gamma \tau)^2, \quad (155)$$

$$\tau^2 \bar{\text{tr}} \Sigma K^{-2} = \bar{\text{tr}} \Sigma (\Sigma + \theta I_d)^{-2} / (\gamma e)^2 = I_{1,2}(\theta) / (\gamma e)^2, \quad (156)$$

where we have used the definition $\theta = \lambda / (\gamma \tau e)$. Thus, u and ω have limiting values u and ω respectively, which solve the system of linear equations

$$u = \psi \gamma \cdot \gamma^{-2} I_{2,2}(\theta) (1 + u) + \psi \gamma \cdot \gamma^{-2} I_{1,2}(\theta) \omega' = \phi I_{2,2}(\theta) (1 + u) + \phi I_{1,2}(\theta) \omega',$$

$$\gamma \omega' = I_{2,2}(\theta) \omega' + \theta^2 I_{1,2}(\theta) (1 + u) = I_{2,2}(\theta) \omega' + \theta^2 I_{1,2}(\theta) (1 + u),$$

where we have used the identity $\phi \gamma = \psi$. These correspond exactly to the equations given in the lemma. This proves the first part.

For the second part, indeed, $\tau = 1 - \eta_0 / \gamma \rightarrow 1$ in the limit $\gamma \rightarrow \infty$, and so $\theta \simeq \lambda / (\gamma e)$ which verifies the equation

$$\theta \simeq \lambda + \lambda \psi \bar{\text{tr}} \Sigma (\gamma e \Sigma + \lambda)^{-1} = \lambda + \phi \cdot \frac{\lambda}{\gamma e} \bar{\text{tr}} \Sigma (\Sigma + \frac{\lambda}{\gamma e} I_d)^{-1} \simeq \lambda + \theta \bar{\text{tr}} \Sigma (\Sigma + \theta I_d)^{-1} / n,$$

i.e $\theta \simeq \lambda + \theta \text{df}_1(\theta) / n$ and $\theta > 0$. By comparing with the equation $\kappa - \lambda = \kappa \text{df}_1(\kappa) / n$ satisfied by $\kappa > 0$ in (7), we conclude $\theta \simeq \kappa$.

Now, the equations (145) become $\omega' = 0$, and $u = \phi I_{2,2}(\kappa) (1 + u)$, i.e

$$u = \frac{\phi I_{2,2}(\kappa)}{1 - \phi I_{2,2}(\kappa)} \simeq \frac{\text{df}_2(\kappa) / n}{1 - \text{df}_2(\kappa) / n},$$

as claimed. $\square$

G.2 Unregularized Limit

Define the following auxiliary quantities

$$\theta := \frac{\lambda}{\gamma \tau e}, \quad \chi := \frac{\lambda}{\tau}, \quad \kappa := \frac{\lambda}{e}. \quad (157)$$

where τ , e , u , and ω are as previously defined in Section 3.2.

Lemma 2. *In the limit $\lambda \rightarrow 0^+$, we have the following analytic formulae*

$$\chi \rightarrow \chi_0 = (1 - \psi)_+ \cdot \gamma \theta_0, \quad (158)$$

$$\kappa \rightarrow \kappa_0 = (\psi - 1)_+ \cdot \theta_0 / \phi, \quad (159)$$

$$\tau \rightarrow \tau_0 = 1 - \eta_0 / \gamma, \quad (160)$$

$$e \rightarrow e_0 = 1 - \phi \eta_0. \quad (161)$$